

DSA (Distributed Services Architecture)

Distributed Services Architecture (DSA) is another open source IoT platform for performing inter-device communication, logic and applications at every layer of the IoT infrastructure. It allows interaction between devices in a decentralised manner, and enables a network engineer to distribute functionality between discrete computing resources. The DSA community is developing a library called 'distributed service links', which allows protocol translation and data integration to and from third party resources. All the modules are lightweight, facilitating DSBroker, multiple DSLinks and client Web applications on the same low-powered devices like Pi, BeagleBone and other computing platforms.

DSA comprises three main components.

- **NodeAPI:** Also called the IoT blueprint, it is regarded as a common open communication method for all DSA modules.
- **DSLink:** Facilitates data exchange with other connected nodes by abstracting field-bus and third party proposed protocols.
- **DsBroker:** Broadcasts the presence and data structures of connected DSLinks to other subscribing DSLinks and DSBrokers.

Features

- Machine to machine authorisation life cycle.
- Allows users to manage subscriptions for listeners and QoS buffers for publishers.
- Implements DSA Query DSL.
- Manages node attributes, node permissions and connections from DSLinks.
- Full hardware integration support, and support for all protocols for IoT connectivity.

Official website: <http://www.iot-dsa.org/>

Thingsboard.io

Thingsboard.io is another 100 per cent open source IoT platform for data collection, processing, visualisation and device management. It supports all standard IoT protocols like CoAP, MQTT and HTTP as well as cloud and on-premise deployments. It builds workflows based on device life cycle events, REST API events, RPC requests, etc.

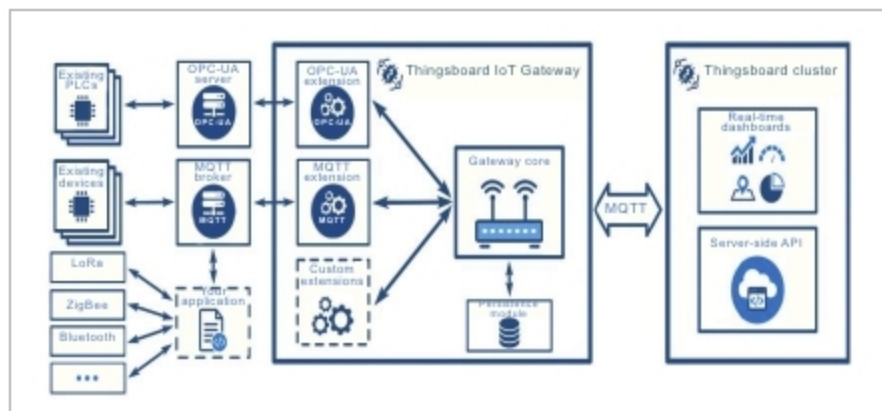


Figure 5: Thingsboard.io

Features

- A strong platform combining scalability, performance and fault-tolerance.
- Easy management of all connected devices in a highly secure manner using rich server side APIs.
- Transforms and normalises device data, and facilitates alarms for triggering alerts on all telemetry events, updates and inactivity.
- Enables use-case specific features using customisable rule chains.
- Handles millions of devices at the same time.
- No single point of failure, as every node in the cluster is identical.
- Multi-tenant installations out-of-the-box.
- Thirty highly customised dashboard widgets for easy user access.

Official website: <https://thingsboard.io>

Thinger.io

Thinger.io is open source and completely hardware agnostic. It provides a ready-to-go platform to design scalable cloud infrastructure that can be fully controlled via an easy GUI (admin) panel and can also be integrated into the business logic with the REST API.

It provides API discovery right out-of-the-box for easy coding and Web interactions. It eliminates the need to select compatible vendor hardware as it provides bloated software to perform all tasks.

Features

- Strong platform for sensing and actuating devices in real-time.
- Countless optimisations under the hood to minimise data bandwidth, processing power and code size; offers server scalability and is fully written in C++ using multi-threaded ASIO for the server side.
- Supports prominent protocols like CoAP and MQTT.
- Highly scalable, secure for hosting data with easy-to-use admin console.
- Easy coding and totally supports modern hardware like Arduino, ESP8266, Raspberry Pi, Intel Edison, etc.
- Docker and IFTTT integration.

Official website: <https://thinger.io/>

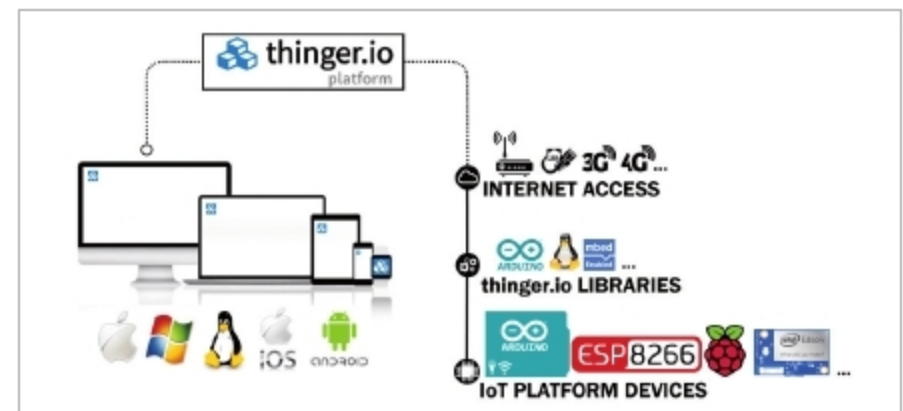


Figure 6: Thinger.io